

Project: Passive RFID Positioning

Team Name: Sniffit

Members: Nader Khalil, Andrew Pang, Victor Porter, Sohan Shah, Braulio Fonseca

Team lead: Nader Khalil

What the project is about?

This project allows the user to pinpoint the location of lost items. The Snapdragon 401 chip, as well as RFID technologies, will be utilized to track an item's location based on RFID data at any time. Losing items is a common worry, and this project can ensure the security of one's items. The theme of security can be extended to beyond just one's belongings: imagine the benefits of knowing the location of any item at any time. Today, this problem is solved with smart devices that can be attached to the item as user wants located (using technologies such as GPS), however, the RFID used is passive (does not require a battery) and thus is far more efficient. Its cost (roughly under 10 cents) makes it a cost-efficient solution too. RFID technology can open many doors since, after this project, without the requirement of a battery, a product can be uniquely identified.

Identify the outcome of the project

At the end of our 20 weeks, we will have a working prototype of a mobile application that allows us to passively track items tagged with a RFID sticker. We want to be able to store location data and create an algorithm that can predict locations where specific items may be lost. On top of that, we will create a user-friendly interface that neatly displays our estimations of where a specific RFID tag is located. In broad terms, we hope to create a low-cost, efficient way to track and notify a user of the location of their lost items.

Define initial project milestones:

Specification:

For specifications, we will first want to verify project details and goals with our mentors. We also want to begin to familiarize ourselves with the technologies and algorithms that would be used for project implementation. This project involves both hardware and software components so getting the team informed on all the technologies will be a crucial part of the specification process. We will also need to design mockups of the networks used and simple interfaces for the front end.

Design:

Determining how the hardware components interact with each other will be the first part of the designing process. After determining a model of interaction between these components, the next phase would be to design an algorithm to calculate location through the RFID readers. On the front end, there will also be a design of a clean interface for users. If time allows, the implementation of augmented reality into the front end could also be part of the design process.

Prototype:

The prototype will be a working demo application that can accurately locate an object in a room. Even though the project specified the use of radio frequencies, the use of other signals such as Wifi and GPS could also be harnessed to determine location in lieu of radio frequency. The prototype will also be heavily tested and debugged.

How do you plan to articulate and design a solution?**List the implementation platform and technologies will plan to use to develop the solution**

UHF RFID Adhesive Tag RFID Label
RFID sensors
Snapdragon 410 board
Debian Linux
Android
LinkSprite UHF RFID Reader Antenna

Overview the process model you will employ to achieve the milestones

We are going to use a scrum framework to guide us through the creation of this project. The twenty weeks before our presentation will be divided into two-week long sprints, with a demo at the end of each sprint. We are using many technologies to assist us in our sprints, such as Slack for general communication, a Github repository to host our code, and Google Drive to hold any documents we are working on and notes on our daily meetings. We also plan to use Trello to define our tasks at the start of each sprint and update progress on each task throughout the sprint.